

Rao Rizwan

Backend Engineer

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SUMMARY

Backend engineer with 2 years building production Python systems. Shipped Django REST Framework APIs serving 100,000+ requests per day, moved long-running work off the request path with Celery and RabbitMQ, and tuned PostgreSQL and MongoDB for heavy throughput. Delivered containerized services on AWS with automated security scanning built into CI/CD.

PROFESSIONAL EXPERIENCE

Broadstone Technologies Jun 2026 – Aug 2026
Backend Developer (Python) Remote

- Built and maintained backend REST APIs in Flask, covering endpoint design, integration, and production support.
- Integrated automated security scanning into CI/CD pipelines, surfacing vulnerabilities and risk findings across application codebases before release.
- Generated AI bill of materials (AIBOM) inventories in-pipeline, giving visibility into AI model and third-party dependency components across projects.
- Automated build, test, and release stages on AWS CodeCommit, CodePipeline, and GitHub Actions, and provisioned EC2, S3, and IAM under least-privilege access.
- Implemented OAuth 2.0 service-to-service authorization across AWS, Azure DevOps, Google Cloud, and GitHub integrations.

Programmers Force Jul 2024 – Apr 2026
Associate Software Engineer Lahore, Pakistan

- Designed and shipped RESTful APIs with Django REST Framework serving more than 100,000 daily requests in production.
- Introduced asynchronous task processing with Celery and RabbitMQ, moving scraping and report generation off the request path onto background workers.
- Tuned the data layer across PostgreSQL and MongoDB through schema design and query optimization to sustain heavy read and write throughput.
- Refactored core API logic and error handling, standardizing response formats and cutting production error rates.
- Containerized services with Docker and Docker Compose, and built resilient web scraping pipelines with BeautifulSoup4 and Selenium.

TECHNICAL SKILLS

Languages: Python, SQL

Backend: Django, Django REST Framework, Flask, FastAPI, REST API design, OAuth 2.0, JWT, Swagger/OpenAPI

Databases: PostgreSQL, MongoDB, Redis, schema design, query optimization

Async: Celery, RabbitMQ, background task processing, scheduled jobs

Cloud & DevOps: AWS (EC2, S3, IAM, CodePipeline, CodeCommit), Docker, Docker Compose, GitHub Actions

Security: Automated vulnerability scanning, software supply chain, AI bill of materials (AIBOM), CycloneDX, Syft

Tools & Platforms: Linux, Git, pytest, Postman, BeautifulSoup4, Selenium

PROJECTS

SiteScopia | Website Analysis Platform [🌐 Live](#) | [🐙 Code](#)

- Built a FastAPI backend that validates and fetches a public URL, parses it into structured facts, and runs independent SEO, accessibility, security, and performance analyzers as background jobs polled by a React frontend.
- Hardened the fetch path against SSRF with URL validation, rate limiting, bounded concurrency, and per-job access tokens; enriched reports via RDAP and DNS. Deployed on Render and Vercel.

(FastAPI, Python, React, Vite)

JobHarvester | Job Aggregation Platform [🌐 Live](#) | [🐙 Code](#)

- Built the Django REST API and PostgreSQL data model behind a live platform serving 1,300+ job listings, with search and filtering by keyword, experience level, and job type.
- Engineered scheduled crawlers with BeautifulSoup4 and Selenium to ingest postings from company career pages, and deployed end to end on Render, Vercel, and Supabase.

(Django REST, PostgreSQL, React)

BOMWatcher | AI Bill of Materials Scanner [🌐 Live](#) | [🐙 Code](#)

- Built a FastAPI service that connects repositories through a GitHub App, opens a pull request to inject a scanning workflow, and ingests the resulting CycloneDX 1.6 artifact over a webhook, so source code never leaves GitHub.
- Paired Syft with a custom AI-model detector to inventory dependencies, hosted AI services and Hugging Face models, flagging copyleft and non-commercial licenses; JWT and Argon2 auth, React 19 frontend, deployed on Vercel and Render.

(FastAPI, SQLAlchemy, React)

EDUCATION

Lahore Garrison University | Bachelor of Software Engineering

Lahore, Pakistan | 2020 – 2024